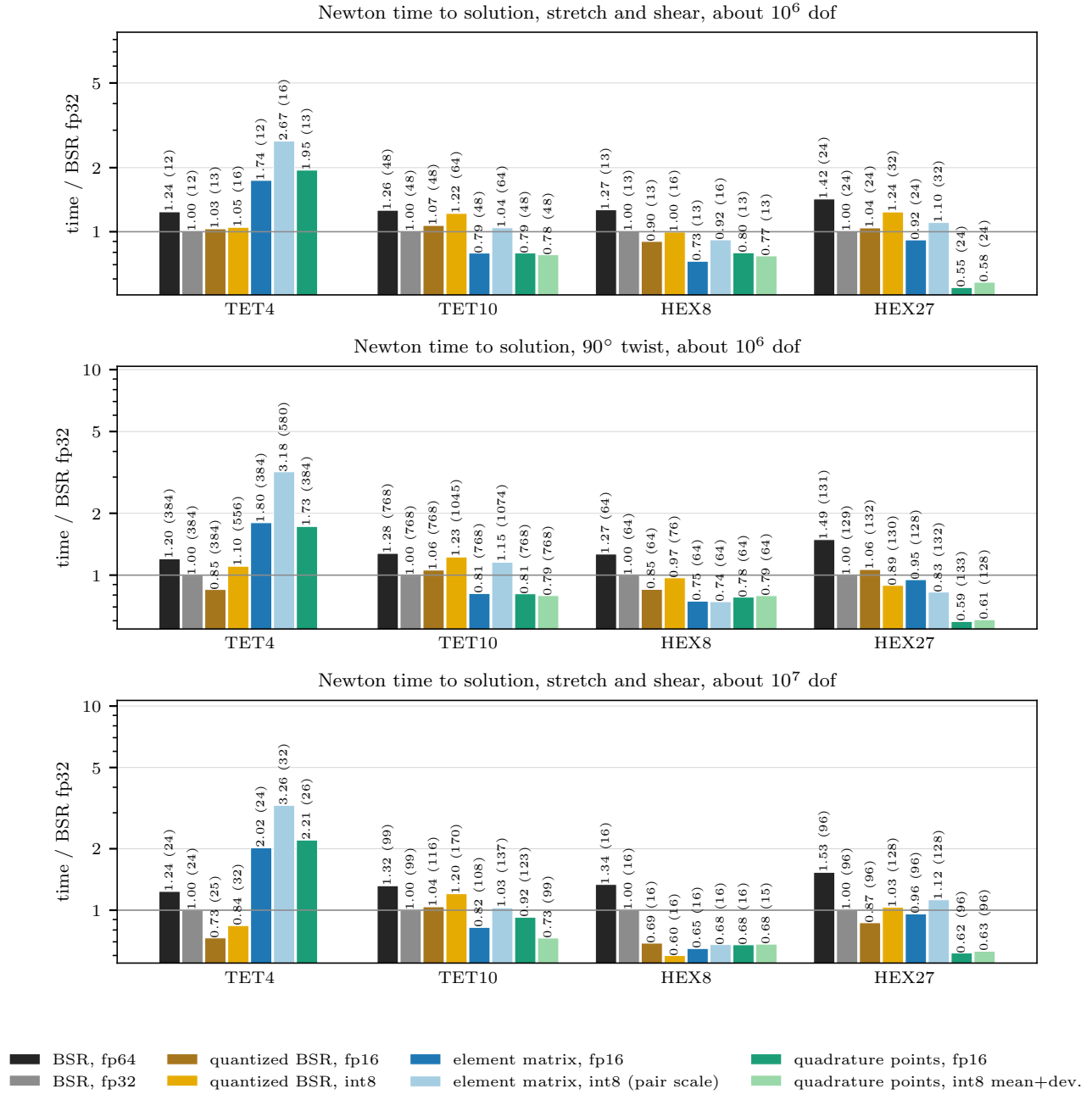

Newton time to solution, stretch and shear, about 10^6 dof
time / BSR fp32
1
2
5
1.24 (12)
1.00 (12)
1.03 (13)
1.05 (16)
1.74 (12)
2.67 (16)
1.95 (13)
1.26 (48)
1.00 (48)
1.07 (48)
1.22 (64)
0.79 (48)
1.04 (64)
0.79 (48)
0.78 (48)
1.27 (13)
1.00 (13)
0.90 (13)
1.00 (16)
0.73 (13)
0.92 (16)
0.80 (13)
0.77 (13)
1.42 (24)
1.00 (24)
1.04 (24)
1.24 (32)
0.92 (24)
1.10 (32)
0.55 (24)
0.58 (24)
TET4
TET10
HEX8
HEX27
Newton time to solution, 90° twist, about 10^6 dof
time / BSR fp32
1
2
5
10
1.20 (384)
1.00 (384)
0.85 (384)
1.10 (556)
1.80 (384)
3.18 (580)
1.73 (384)
1.28 (768)
1.00 (768)
1.06 (768)
1.23 (1045)
0.81 (768)
1.15 (1074)
0.81 (768)
0.79 (768)
1.27 (64)
1.00 (64)
0.85 (64)
0.97 (76)
0.75 (64)
0.74 (64)
0.78 (64)
0.79 (64)
1.49 (131)
1.00 (129)
1.06 (132)
0.89 (130)
0.95 (128)
0.83 (132)
0.59 (133)
0.61 (128)
TET4
TET10
HEX8
HEX27
Newton time to solution, stretch and shear, about 10^7 dof
time / BSR fp32
1
2
5
10
1.24 (24)
1.00 (24)
0.73 (25)
0.84 (32)
2.02 (24)
3.26 (32)
2.21 (26)
1.32 (99)
1.00 (99)
1.04 (116)
1.20 (170)
0.82 (108)
1.03 (137)
0.92 (123)
0.73 (99)
1.34 (16)
1.00 (16)
0.69 (16)
0.60 (16)
0.65 (16)
0.68 (16)
0.68 (16)
0.68 (15)
1.53 (96)
1.00 (96)
0.87 (96)
1.03 (128)
0.96 (96)
1.12 (128)
0.62 (96)
0.63 (96)
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
quantized BSR, fp16
quantized BSR, int8
element matrix, fp16
element matrix, int8 (pair scale)
quadrature points, fp16
quadrature points, int8 mean+dev.